

DIGITAL SIGNAL PROCESSING & SIGNAL ANALYSIS TOOL

Python Source Code

1. main.py

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, filtfilt
import tkinter as tk
from tkinter import filedialog, messagebox

def generate_signal():
    global t, signal
    fs = 1000
    t = np.arange(0, 2, 1/fs)

    clean = np.sin(2 * np.pi * 50 * t)
    noise = 0.5 * np.random.randn(len(t))
    signal = clean + noise

    analyze_signal()

def load_csv():
    global t, signal

    file = filedialog.askopenfilename(
        filetypes=[("CSV Files", "*.csv")]
    )

    if file:
        try:
            data = np.loadtxt(file, delimiter=",", skiprows=1)
            t = data[:, 0]
            signal = data[:, 1]
            analyze_signal()

        except Exception as e:
            messagebox.showerror("Error", str(e))

def analyze_signal():
    fs = 1 / (t[1] - t[0])

    # Low-pass filter
    cutoff = 100
    order = 4

    b, a = butter(order, cutoff / (fs / 2), btype="low")
    filtered = filtfilt(b, a, signal)

    # RMS calculation
    rms = np.sqrt(np.mean(filtered ** 2))

    # FFT calculation
    n = len(filtered)
    fft_result = np.fft.fft(filtered)
    frequencies = np.fft.fftfreq(n, 1 / fs)

    positive = frequencies >= 0
    magnitude = np.abs(fft_result) / n

    freq = frequencies[positive]
    mag = magnitude[positive]

    # Dominant frequency
    dominant_frequency = freq[np.argmax(mag)]

    result_label.config(
        text=f"RMS Value: {rms:.3f}\n"
        f"Dominant Frequency: {dominant_frequency:.2f} Hz"
    )

    # Graphs
    plt.figure(figsize=(10, 7))
```

```

plt.subplot(2, 1, 1)
plt.plot(t, signal, label="Original Signal")
plt.plot(t, filtered, label="Filtered Signal")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.title("Time Domain Signal")
plt.legend()
plt.grid()

plt.subplot(2, 1, 2)
plt.plot(freq, mag)
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.title("Frequency Spectrum using FFT")
plt.xlim(0, 200)
plt.grid()

plt.tight_layout()
plt.show()

# GUI
root = tk.Tk()
root.title("Digital Signal Processing & Analysis Tool")
root.geometry("500x350")

title = tk.Label(
    root,
    text="Digital Signal Processing Tool",
    font=("Arial", 18, "bold")
)
title.pack(pady=25)

generate_btn = tk.Button(
    root,
    text="Generate Sample Signal",
    command=generate_signal,
    width=25,
    height=2
)
generate_btn.pack(pady=10)

load_btn = tk.Button(
    root,
    text="Load Signal from CSV",
    command=load_csv,
    width=25,
    height=2
)
load_btn.pack(pady=10)

result_label = tk.Label(
    root,
    text="RMS Value: --\nDominant Frequency: --",
    font=("Arial", 12)
)
result_label.pack(pady=25)

root.mainloop()

```

2. requirements.txt

```

numpy
scipy
matplotlib

```

3. Installation & Execution

Install the required libraries using: **pip install numpy scipy matplotlib**

Run the project using: **python main.py**

4. CSV Input Format

```

Time,Amplitude
0.000,0.12
0.001,0.25
0.002,0.48

```

0.003, 0.71